Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	***Autonomous Kingpin Coupling System with 3D Vision-Guided Robotic Arm for Heavy-Load Trailer Automation***			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility (○) Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Machinery and materials		*Machinery and materials*	
	Medium Category	Small Category	Medium Category	Small Category
	Automobile / Rail Vehicles	*Vehicle Intelligence Technologies*	*Automobile / Rail Vehicles*	*Low-Emission Vehicle Technologies*
Concept of Proposed Technology	**Core Concept & Components** This technology enables **fully autonomous coupling between semi-trailers and autonomous yard tractors** without human intervention, specifically designed for smart factory yard automation. 1. **3D Vision System**: Stereo camera + structured light hybrid sensor for precise kingpin detection (±2mm accuracy) 2. **6-DOF Robotic Arm**: Automated connection and guidance of kingpin, air brake lines (red/blue glad hands), and electrical connector 3. **AI-Powered Control**: Real-time pose estimation, adaptive force control, and self-learning algorithms 4. **Safety Mechanisms**: Multi-layer collision avoidance, force/torque feedback, and emergency override **Main Functions:** • Autonomous detection and alignment to trailer kingpin (50mm or 90mm standard)			

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	• Automatic coupling/decoupling cycle completion in 40-60 seconds • Zero trailer modification required - works with existing fleet • Operation in mixed traffic yards with 98%+ success rate **Key Functions and Potential Applications** • Smart factory yard automation • Port/terminal container trailer handling • Logistics center trailer management • Hazardous material handling environments
Rationale for Joint R&D Project	**Market Necessity:** • Korea's aging truck driver population (avg. 56 years) requires automation solutions • Smart factory market growing at 15-20% CAGR • Global autonomous yard truck market reaching $2B+ by 2030 **Why Global Collaboration?** 1. **Complementary Expertise**: - Foreign Partner: 3D vision algorithms, robotics control systems, AI/ML models Purdue University, a leading overseas research institution, is a manufacturing-focused engineering university located in Indiana, a region surrounded by numerous manufacturing companies, and serves as an industrial hub in the United States. The university has already been collecting meaningful equipment data for smart manufacturing and digital transformation in the surrounding area and utilizing them for AI-based research. Purdue maintains strong collaborative relationships with medium- to large-scale enterprises equipped with factories, facilities, and production lines; it operates specialized research centers capable of conducting testbed-based research on manufacturing and physical AI; it is strategically located in a major transportation hub in the U.S., making it the only academic institution where Korean companies can directly identify technological trends and market demands in the local environment. Therefore, global collaboration is essential. - Korean SME: Automotive mechanical systems, 5th wheel coupling mechanisms, local market requirements 2. **Localization Requirements**: - Korean trailer specifications (KS standards) - Integration with domestic YMS/WMS platforms - Compliance with Korean safety regulations 3. **Cost Optimization**: - Joint development reduces R&D investment by 30-40% - Shared testing infrastructure and validation facilities - Faster time-to-market through parallel development 4. **Technology Transfer**: - Korean SME gains cutting-edge robotics/vision expertise - Foreign partner gains access to Asia-Pacific automotive supply chain - Both parties benefit from IP co-ownership
Key Words	1. **Kingpin Automatic Coupling**

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	2. **3D Vision-Guided Robotics** 3. **Autonomous Yard Tractor** 4. **Sensor Fusion** 5. **Smart Factory Automation**

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (4)	**End of the Project**	Level (7)
R&D Objectives	**Primary Objective**: Develop a fully autonomous kingpin coupling system capable of 40-60 second cycle time with 98%+ success rate in real-world yard environments **Specific Technology Development Goals:** **1. Advanced 3D Vision System** • Develop hybrid depth sensing (stereo + structured light) with ±2mm position accuracy • AI-based kingpin detection robust to lighting conditions (0-100K lux) • Real-time pose estimation at 50Hz update rate • Operation in adverse weather (rain, snow, fog) **2. Precision Robotic Coupling Mechanism** • Design 6-DOF collaborative robot arm with 1200mm reach, ±0.5mm repeatability • Custom end-effector for glad hand (air line) and 7-way (electrical) connection • Force-controlled insertion with <20N contact force • Automatic verification of connection integrity **3. Intelligent Control Algorithms** • Multi-sensor fusion (vision + IMU + encoders) using Extended Kalman Filter • Adaptive PID control with learning-based optimization • Collision avoidance with 360° LiDAR integration • Fail-safe mechanisms for Minimal Risk Condition (MRC) **4. System Integration & Safety** • Redundant control architecture • Communication interface with Yard Management Systems (YMS) • Remote monitoring and teleoperation capability • Emergency override and manual backup procedures			
R&D Target Specifications	**Performance Specifications:** **Vision System** • Kingpin detection range: 0.5-3.0 meters • Position accuracy: ±2mm (X, Y, Z axes) • Orientation accuracy: ±0.5° (roll, pitch, yaw) • Detection success rate: >99.5% • Processing latency: <50ms per frame • Operating temperature: -20°C to +50°C **Robotic Arm** • Degrees of freedom: 6-axis			

	• Payload capacity: 15 kg • Reach: 1200 mm • Repeatability: ±0.5 mm • Maximum velocity: 250 mm/s (approach), 50 mm/s (coupling) • Force sensing resolution: 0.1 N **Coupling Performance** • Total cycle time: 40-60 seconds (approach + couple + verify) • Kingpin coupling time: <15 seconds • Air line connection time: <20 seconds per line (2 lines total) • Electrical connection time: <10 seconds • Success rate: ≥98% (first attempt) • Retry success rate: ≥99.5% (with realignment) **Safety & Reliability** • Mean Time Between Failures (MTBF): >1,000 hours • Emergency stop response: <100ms • Collision detection distance: 0.5 meters • Redundancy level: ISO 13849 Category 3 • Operating cycles before maintenance: 10,000 cycles **Environmental** • Dust/water resistance: IP67 • Vibration tolerance: 5G (IEC 60068-2-6) • Wind resistance: 15 m/s • Acoustic emission: <75 dBA at 1 meter
R&D Plan	**42-Month Development Plan** **Phase 1: Foundation (Months 1-6, TRL 4→5)** **Vision System Development** • Month 1-2: Hardware selection and procurement - Stereo camera system (baseline 200mm, 2MP each) - Structured light projector (850nm IR, coded pattern) - Computing platform (NVIDIA Jetson AGX Orin) • Month 3-4: Algorithm development - Point cloud preprocessing and filtering - Circular feature detection (Hough Transform) - 3D pose estimation (PnP + RANSAC) • Month 5-6: Laboratory validation - Static kingpin detection tests (100 samples) - Performance characterization (accuracy, latency) **Robotic System Design** • Month 1-3: Mechanical design - Kinematic analysis and optimization - Structural FEA (Finite Element Analysis) - End-effector design for glad hand/electrical connector • Month 4-6: Prototype fabrication and testing - Arm assembly and integration - Joint calibration and characterization - Basic motion control implementation

Milestone 1 (Month 6): Vision system detects kingpin with 95% success rate; robotic arm completes basic motion sequences

Phase 2: Integration (Months 7-18, TRL 5→6)

System Integration
• Month 7-9: Hardware-software integration
- ROS2-based control architecture
- Vision-robot coordinate transformation
- Force/torque sensor integration
• Month 10-12: Control algorithm development
- Sensor fusion (EKF implementation)
- Visual servoing for alignment
- Impedance control for coupling
• Month 13-15: Glad hand connection development
- Gripper mechanism design
- Rotation and locking sequence
- Air pressure verification
• Month 16-18: Electrical connector automation
- 7-way plug alignment mechanism
- Insertion force control
- Continuity testing

Indoor Testing
• Month 12-18: Controlled environment tests
- 500+ coupling cycles with static trailer
- Variety of kingpin conditions (clean, dirty, rusty)
- Different lighting scenarios

Milestone 2 (Month 18): Complete autonomous coupling cycle achieved in laboratory; 90% success rate

Phase 3: Validation (Months 19-30, TRL 6→7)

Outdoor Testing
• Month 19-21: Test facility setup
- Outdoor yard environment preparation
- Multiple trailer types (2" and 3.5" kingpins)
- Weather monitoring equipment
• Month 22-27: Performance validation
- 2,000+ coupling cycles in realistic conditions
- Day/night operation testing
- Rain, fog, and temperature extremes
- Moving trailer scenarios (wind sway)
• Month 28-30: Reliability testing
- Continuous operation tests (100 cycles/day)
- Failure mode analysis
- Maintenance procedure development

AI Enhancement
• Month 24-30: Machine learning optimization
- Data collection (10,000+ images)
- CNN-based kingpin detector training

	- Reinforcement learning for coupling strategy - Model deployment and edge optimization **Milestone 3 (Month 30)**: System achieves 98%+ success rate in outdoor testing; AI model improves detection robustness by 15% **Phase 4: Pilot Deployment (Months 31-42, TRL 7→8)** **Customer Site Integration** • Month 31-33: Site preparation - YMS interface development - Network infrastructure setup - Safety protocol implementation • Month 34-39: Pilot operation - Initial supervised operation (500 cycles) - Gradual autonomy increase - 24/7 operation testing (2,000+ cycles) - Operator training and feedback • Month 40-42: Commercialization preparation - Design for Manufacturing (DFM) optimization - Cost reduction analysis (target: 30% reduction) - Certification preparation (ISO, CE, KC) - User manual and training materials **Safety Certification** • Month 36-42: Standards compliance - ISO 3691-4 (Autonomous industrial vehicles) - ISO 13849-1 (Safety control systems) - Korean certification (KC, KCs) - Risk assessment (FMEA, FTA) **Milestone 4 (Month 42)**: Pilot site operation demonstrates commercial viability; TRL 7 achieved; ready for production scale-up **Key Deliverables:** ✓ Fully functional prototype system (3 units) ✓ Technical documentation (design, software, testing) ✓ Performance validation report (>5,000 cycles) ✓ Safety certification documentation ✓ Manufacturing drawings and BOM ✓ Training materials and operation manual ✓ Patent applications (minimum 3) ✓ Research publications (minimum 2)

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute (Purdue University), led by the overseas partner, will be responsible for designing Robot control, algorithms and AI models, and developing the motion planning and integrations. **1. Core Technology Development**

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	• 3D Vision Algorithms - Stereo vision and structured light fusion algorithms - AI/ML models for kingpin detection and pose estimation - Real-time image processing optimization - Robustness enhancement for adverse conditions • Robotics Control Systems - 6-DOF arm kinematic and dynamic modeling - Motion planning and trajectory optimization - Force/torque control algorithms - Collision avoidance and safety mechanisms • System Integration - Sensor fusion framework (vision + IMU + encoders) - ROS2-based software architecture - Communication protocols with external systems **2. Research Scope** • Academic research on advanced coupling strategies • Algorithm development and simulation (MATLAB/Gazebo) • Prototype software implementation and testing • AI model training infrastructure and datasets • Performance benchmarking and optimization **3. Resources to be Provided** • Research personnel (5-8 researchers/engineers) • Computing infrastructure (GPU servers, cloud resources) • Simulation and testing software licenses • Laboratory testing facilities • Technical documentation and know-how **4. Specific Tasks** • Phase 1: Vision system algorithm development and validation • Phase 2: Robotic control system design and integration • Phase 3: AI-based optimization and learning algorithms • Phase 4: Technical consulting for pilot deployment • Ongoing: IP management, publication support, training
Expected R&R of Korean SME Partner	The Korean SME partner will take primary responsibility for end-to-end mechanical system development, prototyping and manufacturing, testing and validation, localization, regulatory compliance, and commercialization in the Korean market. **1. Mechanical Systems Development** • 5th Wheel Coupling Expertise - Korean trailer specifications and standards (KS) - Kingpin mechanical interface design - Coupling mechanism durability testing • Robotic Arm Hardware - Mechanical design and fabrication - End-effector custom development - Structural optimization for automotive environment • Vehicle Integration - Mounting system design for yard tractors - Electrical/pneumatic system integration - Compliance with vehicle safety regulations

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	2. Development Tasks • Hardware prototyping and manufacturing • Component sourcing from Korean automotive supply chain • Cost engineering and value analysis • Design for Manufacturing (DFM) optimization • Production scale-up planning **3. Testing and Validation** • Field testing coordination at customer sites • Environmental testing (temperature, humidity, dust) • Durability and reliability testing (10,000+ cycles) • Integration with Korean YMS/WMS platforms • User acceptance testing with Korean fleet operators **4. Resource Commitments** • Engineering team (8-12 engineers: mechanical, electrical, software) • Manufacturing facilities for prototyping • Testing equipment and outdoor yard facility • Customer relationship and pilot site coordination • Local certification and regulatory compliance support **5. Collaboration Efforts** • Weekly technical coordination meetings • Joint testing and validation activities • Shared data collection and analysis • Co-authorship on patents and publications • Market research and commercialization strategy • After-sales service and maintenance infrastructure **6. Localization Activities** • Adaptation to Korean market requirements • Korean language user interface development • Integration with domestic logistics software • Sales and marketing in Korea/Asia-Pacific • Customer training and technical support

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technological Achievements and Intellectual Property** The project will develop a world-class AI- and 3D vision-based robotic coupling system with 99.5%+ detection accuracy, <20N force-controlled connection, hybrid sensor fusion, and zero-modification compatibility. It will secure 3–5 patents, build a proprietary software stack, and accumulate core technical know-how. Key performance targets include 40–60 second cycle time, ≥98% success rate, -20°C to +50°C operation, and >1,000 hours MTBF. **Commercialization Potential** The project targets a global autonomous yard truck market of $2.5B by 2030 and a Korean smart factory market of ₩50T by 2027, covering 50,000+ domestic and 500,000+ global units. The roadmap includes certification and initial sales in Years 1–2 (10–20 units), mass production in Year 3 (100–200 units), and global expansion in Years 4–5 (500+ units/year). Revenue targets include ₩50–80M per unit, ₩10–20B over five years, and ₩2–4B from aftermarket services. The

	business model combines direct sales, retrofit kits, SaaS services, and technology licensing. **Industrial, Economic, and Social Impact** The technology will enable 24/7 unmanned operations, reduce accidents by 90%, and improve throughput by 30–50%, addressing aging driver shortages. It will create 50–100 jobs, generate ₩30–50B in annual SME procurement, and strengthen exports and regional economies. Technology spillover will advance robotics, AI, and smart factory infrastructure and support autonomous construction and mining equipment. **Global Competitiveness and R&D Ecosystem** The project will position Korea as a leader in yard automation, compete with global players, establish national standards, and strengthen international partnerships. It will enhance university-industry collaboration, train next-generation engineers, and build testing and validation infrastructure. **Success Metrics (5 Years Post-Project)** Within five years, the project aims to deploy 300+ systems, achieve 15–20% domestic and 5–8% global market share, exceed 85% customer satisfaction, rank in the global top three for speed and reliability, generate ₩150B+ revenue and ₩30B+ exports, secure 10+ patents, and contribute to ISO/SAE standards.

6. References

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(2) Technical Publications and Patents

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- Outrider Technologies (2025). "Reinforcement Learning for Autonomous Yard Operations," Technical White Paper
- US Patent US9327782B2: "Fifth wheel locking mechanism with automated sensor verification"
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(3) Academic Research

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- Vision-based Trailer Pose Estimation for Articulated Vehicles, ResearchGate
- Model Predictive Path-Following Control for Truck-Trailer Systems, ScienceDirect (2024)

(4) Industry Standards

- ISO 4086:2001: Road vehicles — 90 semi-trailer fifth wheel kingpin — Interchangeability
- ISO 337: Fifth wheel coupling — 2" (50mm) kingpin specifications
- ISO 3691-4: Industrial trucks — Safety requirements — Autonomous industrial trucks
- ISO 13849-1: Safety of machinery — Safety-related parts of control systems
- SAE J560: 7-way electrical connector standard

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(5) Technology Benchmarks

- JOST International: JSK37UBK Sensor Fifth Wheel Coupling Technology
- ZF Group: Automated Coupling Assist System
- SAF-HOLLAND: RECOSS Safety Sensor System
- Stoughton Trailers: Auto Connect™ System

(6) Supporting Organizations

- TIPA (Technology and Information Promotion Agency for SMEs)
- Korea Automotive Technology Institute (KATECH)
- Korea Institute of Robotics and Technology Convergence (KIRO)
- Korea Transportation Safety Authority (TS)